

Examination	University	Institute	CGPA/Percentage
BTech	Anna University	Karpagam College of Engineering	8.5*
HSC	CBSE	Shri Maharishi Vidya Mandir	81%
SSLC	CBSE	Shri Maharishi Vidya Mandir	81%

SKILL SUMMARY

Languages: Java, Python, SQL, Html, CSS

Frameworks: Flask, Spring Boot, Java Servlets, JSP, scikit-learn, TensorFlow, PyTorch, pandas, Kafka, Matplotlib, Hugging face.

Developer Tools: Git, GitHub, Jupyter Notebook, VS Code, Eclipse, Docker, Streamlit, Kaggle.

Visualization Tools & Software: Power Bi, Tableau, Excel, Python visualization.

Web Development: Html, css, Servlet, Jsp, Fast API, Streamlit

Soft Skills: Problem Solving, Analytical Thinking, Clear Communication, Adaptability, Team Collaboration, Presentation

KEY PROJECTS

- Uzhavar Kaappaan** | *Python, Flask, AI/ML, NLP, Speech Recognition, Twilio, Route Optimization, AI Agents* [Feb '26]
- Created a voice-driven AI agriculture platform that converts farmers' natural language voice inputs into intelligent, actionable advisories.
 - Automated speech understanding, intent extraction, and decision reasoning, including crop care, irrigation, pest control, and yield planning.
 - Enabled direct produce sales to nearby shops through automatic AI voice calls, allowing farmers to connect with buyers easily without middlemen.
- TraceChild** | *Deep Learning, Computer Vision, Face Recognition, Age Progression GANs, CNNs, OpenCV* [June '25]
- Created an AI-driven missing child identification system that uses age progression to assist in locating children years after disappearance.
 - Applied deep learning-based face analysis and age progression models to predict current appearances from historical images.
 - Automated face matching and similarity scoring against updated image databases to support faster and more accurate identification.
- AI-Powered B2B Data Quality Guardian** | *Multi-Agent AI Architecture, Data Validation, JSON Reporting* [Dec '25]
- A real-time AI-powered data quality guardian that continuously monitors enterprise datasets, detects quality issues, and assesses data reliability using rule-based and statistical validation.
 - Designed a multi-agent digital twin of data pipelines to continuously simulate, monitor, and validate data flow, schema changes, and anomalies in real time without human intervention.
- Multimodal Voice RAG Assistant** | *Python, AI Agents, Streamlit, NVIDIA Nemotron, FAISS, RAG, LangChain* [Jan '26]
- Created a voice-driven AI assistant enabling natural speech and text interaction in a ChatGPT-style UI. Applied computer vision-based pose estimation and activity recognition to monitor exercise form, repetitions, and workout consistency in real time.
 - Implemented RAG-based knowledge retrieval using FAISS and Nemotron embeddings for accurate, context-aware responses.
 - Integrated speech transcription, safety guardrails, and dynamic document indexing for real-time knowledge updates.
- EcoPulse AI** | *Real-Time Streaming Analytics, Kafka, Pathway, Predictive Modeling, Flask* [Jan '26]
- Built a real-time environmental intelligence platform using Kafka and Pathway to stream and analyze AQI data with anomaly detection.
 - Implemented predictive risk forecasting and AI-driven action recommendations to support proactive environmental decision-making.

SCHOLASTIC ACHIEVEMENTS

- Participated in AIMs '25, an industry-level AI/ML hackathon organized by Infynd, gaining exposure to real-world problem statements [Dec '25]
- Participated in Ambiton'25, a 24 hours hackathon at Bangalore Institute of Technology [Sept '25]
- Presented a paper on Evaluating Groundwater Quality in Tirupur: Hydrochemical Facies - CIT [Feb '25]

CERTIFICATES

Responsible & Safe AI Systems
Swyam NPTEL - 61%

RDBMS PostgreSQL
Spoken Tutorial

MongoDB Basics for Students
credly

Introduction to IOT
Swyam NPTEL – 75%

Enhancing Soft Skills and Personality
Swyam NPTEL – 75%